AI Applications in Research: The Use of Claude Code

Concept summary

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Abstract

This document summarizes the main concepts around applying agentic AI to research work, with an emphasis on **Claude Code**. It draws on the practical, first-person experience reported in Scott Cunningham's *Scott's Mixtape Substack* (an economist adopting AI agents for empirical causal-inference work) and on general practice for AI-assisted research. The themes are organized as: (i) what agentic coding tools are; (ii) how they extend to external systems via connectors and the Model Context Protocol (MCP); (iii) the emerging discipline of a research "harness"; and (iv) the recurring cautions of *production, verification, and understanding.*

1 Why AI Agents in Research

Traditional research computing has the researcher writing code directly in a text editor, reading and seeing every line. **Agentic AI** tools invert this: the researcher delegates implementation to a model that reads the project, plans, edits files, runs code, and reports back. The promise is a large increase in **labor productivity**—one person can accomplish what previously required a team—while the risk is a loss of direct visibility into what was actually done.

Cunningham frames the productivity gain in explicitly economic terms: as a single worker produces more output (Y/N), that output feeds per-capita income (Y/Pop) and, through market mechanisms, both growth ("the wealth of nations") and distribution ("the wages of nations"). A concrete signal is the rise of **solo-entrepreneurs** ("solopreneurs")—very small firms, sometimes a single employee, an increasing share of which report \$1–10M in annual revenue—a phenomenon partly enabled by AI tooling and payment/back-office integrations.

2 Claude Code: The Agentic Coding Environment

Claude Code is a command-line agent that operates inside a project. Its core research-relevant capabilities are:

- **Whole-project context**: it reads the repository, not just a single snippet, so it can navigate a real analysis codebase.
- **Tool use**: it runs shell commands, executes scripts, edits files, and inspects outputs in a loop.
- **Reproducibility discipline**: good practice (echoed by Cunningham) is to *never edit raw data* in place and to route all transformation through inspectable, re-runnable code.
- **Hooks**: user-defined scripts that fire on agent events (e.g. before/after a tool call) to enforce guardrails—formatting, validation, or protecting raw data—so the workflow encodes the researcher's standards automatically.

- **Availability beyond the laptop**: the same agentic capability is increasingly reachable from the Claude app (including on a phone) and from cloud environments, not only the local machine.

3 Connectors and the Model Context Protocol (MCP)

A central late-2026 theme is **connectors**: integrations that let the assistant reach out to external data sources and services and take actions in them. Key points:

- **MCP (Model Context Protocol)** is the open standard underneath connectors—"a new kind of piping" that lets an AI application securely connect to external tools and data and perform actions in real time.
- The connector directory functions like an "app store," on the order of *thousands* of entries and growing, spanning categories such as *Code*, *Financial Services*, and productivity tools.
- **First-party vs. Community connectors**: some are published by the vendor; *Community* connectors are user-created and only lightly vetted—adopt them with a "buyer beware" posture, especially where credentials or sensitive data are involved.
- **Business/back-office reach**: connectors for Stripe, PayPal, QuickBooks, Gusto (payroll), and Dropbox make it feasible for one person to run substantial operations from the chat interface; banking APIs (e.g. direct bank access) remain conspicuously absent, plausibly for security reasons.
- **Research relevance is still emerging**: Cunningham is candid that the payoff of connectors for *empirical research* (as opposed to running a small business) is not yet obvious to him, but he expects some cloud-only resources will eventually be reachable efficiently *only* through a connector.

4 The "Harness": Staging Research Cognition

The most distinctive research concept in Cunningham's writing is the **harness**—not a single tool but a structured *staging of the researcher's own cognition* so that projects move through disciplined stages. Its design principles:

- **Checklist philosophy**: each stage maps to a checklist. This mirrors the end-of-chapter "Practical Tips / Practical Advice / Concluding Remarks" distilled throughout his book *Causal Inference: The Remix*, and the design-based tradition he credits to Don Rubin, Guido Imbens, and Pedro Sant'Anna.
- **Design principles for causal inference**: the harness is meant to *discipline the mind*—forcing the analyst to state the identification strategy, covariate choices, and assumptions *before* trusting an estimate.
- **Visualization to restore "seeing"**: because delegating code to an agent removes the direct sight of one's own code, the harness re-introduces visual structure—interactive, "flippable card" figures and a dashboard—so the researcher can remember and review what has been done. Cunningham even imagines mapping the harness into an Apple Vision Pro spatial interface.
- **Connector-supported, not connector-defined**: he expects the eventual harness to *use* connectors heavily but to remain a graphical, workflow-centered environment of its own.

5 The Governing Triad: Production, Verification, Understanding

Cunningham reduces the goal of AI-assisted research to three requirements that serve as a useful rubric for any researcher adopting these tools:

1. **Production**—actually generating the analysis (estimates, tables, figures) efficiently.
2. **Verification**—confirming the output is what the researcher intended *and* that it does what it claims to do (the estimator is correctly specified, the data pipeline is sound, results replicate).
3. **Understanding**—maintaining a continuous, human grasp of what has been done, so the researcher is not merely accepting agent output on trust.

The honest tension he reports—"drinking from a firehose" and having to "wrestle" the agent to work on his terms—is itself an important concept: raw agent speed is not the same as trustworthy research, and the whole point of the harness and the triad is to convert capability into *credible*, inspectable scholarship.

6 Practical Takeaways for Researchers

- Treat the agent as a fast collaborator, not an oracle: keep raw data immutable, route everything through re-runnable code, and read the diffs.
- Encode your standards mechanically (hooks, checklists) rather than relying on remembering them each session.
- Adopt connectors deliberately; prefer vetted ones and scrutinize Community connectors before granting access to sensitive systems.
- Build (or borrow) a lightweight "harness": explicit stages, a verification step, and a visual record of what was produced.
- Judge every workflow against production, verification, and understanding—if any one is missing, the result is not yet research.

Sources

- Scott Cunningham, "Connectors and My Harness Idea," *Scott's Mixtape Substack*, 9 Sept. 2026. https://causalinf.substack.com/p/connectors-and-my-harness-idea
- Related recent posts (same publication, Aug.–Sept. 2026): "How and why I am using hooks," "Some thoughts about AI in practice," "Where and how will the human be in the loop?," and "Memory is the residue of thought."
- Scott's Mixtape Substack: https://causalinf.substack.com/
- Scott Cunningham, *Causal Inference: The Remix* (Yale University Press, 2026).